

CS675

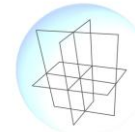
Computer Graphics

Introduction

Parag Chaudhuri



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IIT Bombay



ViGIL
VISION, GRAPHICS,
IMAGING & LEARNING

Introductions



Parag Chaudhuri

<https://www.cse.iitb.ac.in/~paragc>

Introductions

- Registered - 100
 - BTech(73), Dual(4), MTech(15), MS(6), PhD(2)
- Waitlisted - 75

What is this course all about?

What is Computer Graphics ?



What is Computer Graphics used for ?



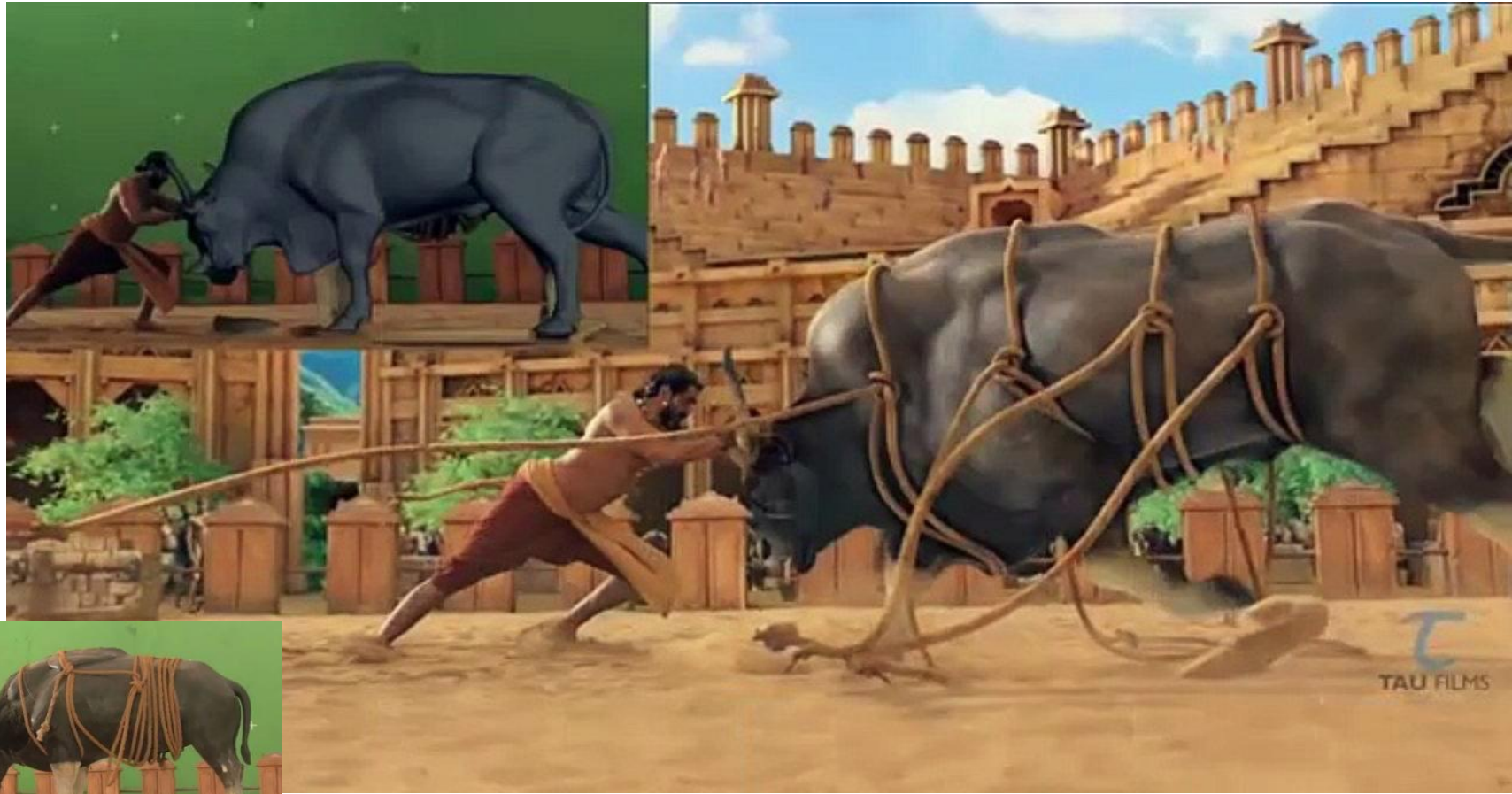


Cinematic Animation



Ernest and Celestine, Les Armateurs, 2013

VFX



Bahubali 2, Arka Mediaworks and Tau Films, 2015

VFX



The Jungle Book, Disney, 2016

Computer Games

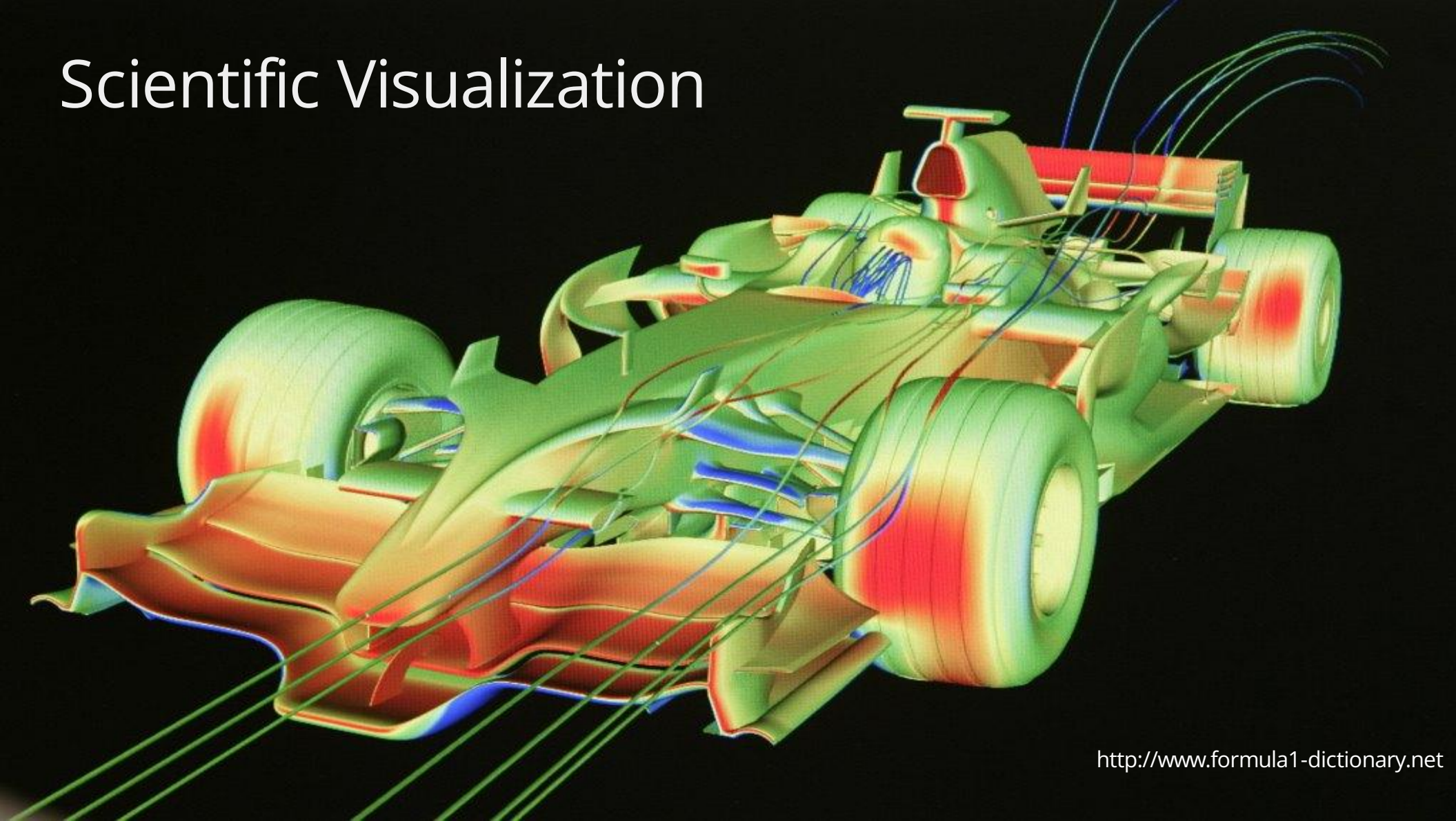


Age of Empires 3, Ensemble Studios, 2005

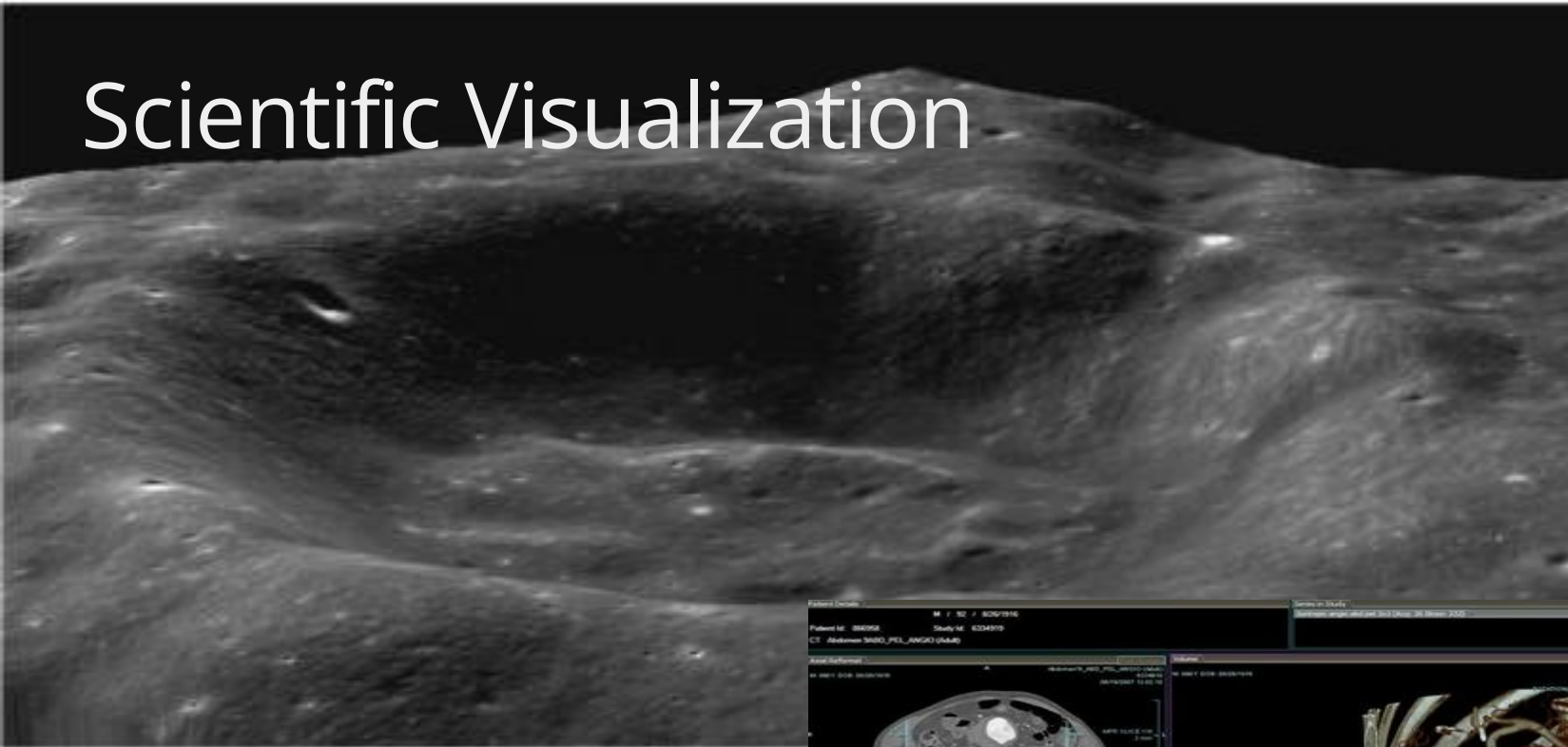
AGE OF EMPIRES 3
ENSEMBLE STUDIOS

A 3D architectural rendering of a complex building structure. The central part of the building is highlighted in a vibrant green color, showing a multi-level interior with a grid-like floor plan and a central staircase. This green volume is surrounded by other white, 3D-modeled building sections, including a large rectangular block and a more complex, multi-story structure with a grid-like facade. The entire 3D model is set against a background of various architectural blueprints and floor plans, which include dimensions, room labels, and structural lines. The text 'CAD' is prominently displayed in the top left corner in a blue, sans-serif font. The overall composition suggests a professional architectural design process using CAD software.

Scientific Visualization

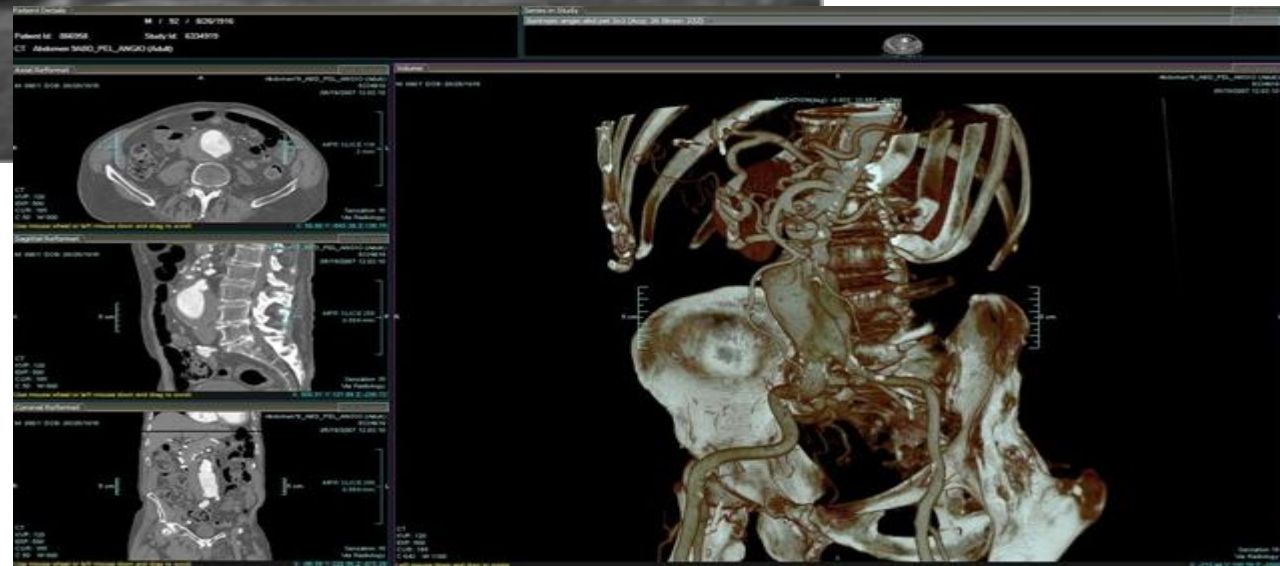


Scientific Visualization



Fiatlux Imaging

Chandrayaan, ISRO



Augmented and Virtual Reality



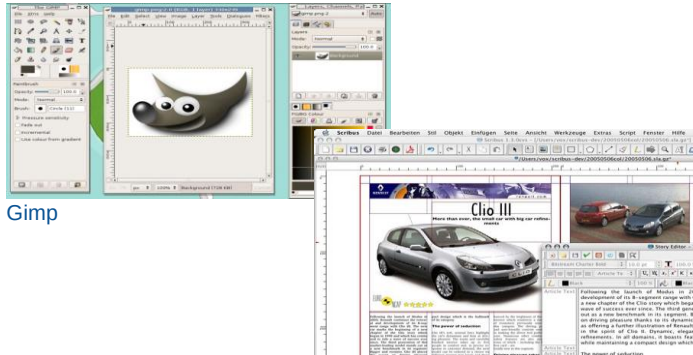
Microsoft HoloLens

Digital and Print Media



Computational Photography

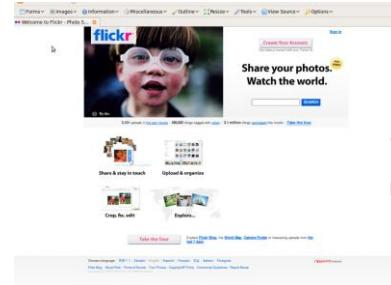
A Agarwala, M Dontcheva, M Agrawala, S Drucker, A Colburn, B Curless, D Salesin, M Cohen:
Interactive Digital Photomontage. ACM Transactions on Graphics (Proceedings of SIGGRAPH 2004)



Gimp

Scribus

Desktop Publishing



Networked Media

Flash, Silverlight, HTML5 Canvas
IE, Firefox, Safari, Opera
YouTube, Flickr, Facebook

Graphical User Interfaces



Unity/Ubuntu



Aqua/Mac MacOS



Metro/Windows 10

Now

Microsoft
Surface Hub



Wacom
Cintiq

Ivan Sutherland's
Sketchpad



Apple iPhone



The Future?



© Twentieth Century Fox

Fraunhofer FIT



1962

And the list goes on...

- Virtual Worlds – Google Earth, Second Life, Minecraft
- Algorithm Animation
- Typesetting and font design – LaTeX, Freetype
- Digital Video and HDTV
- Drug Design
- Image and Video Search
- ...

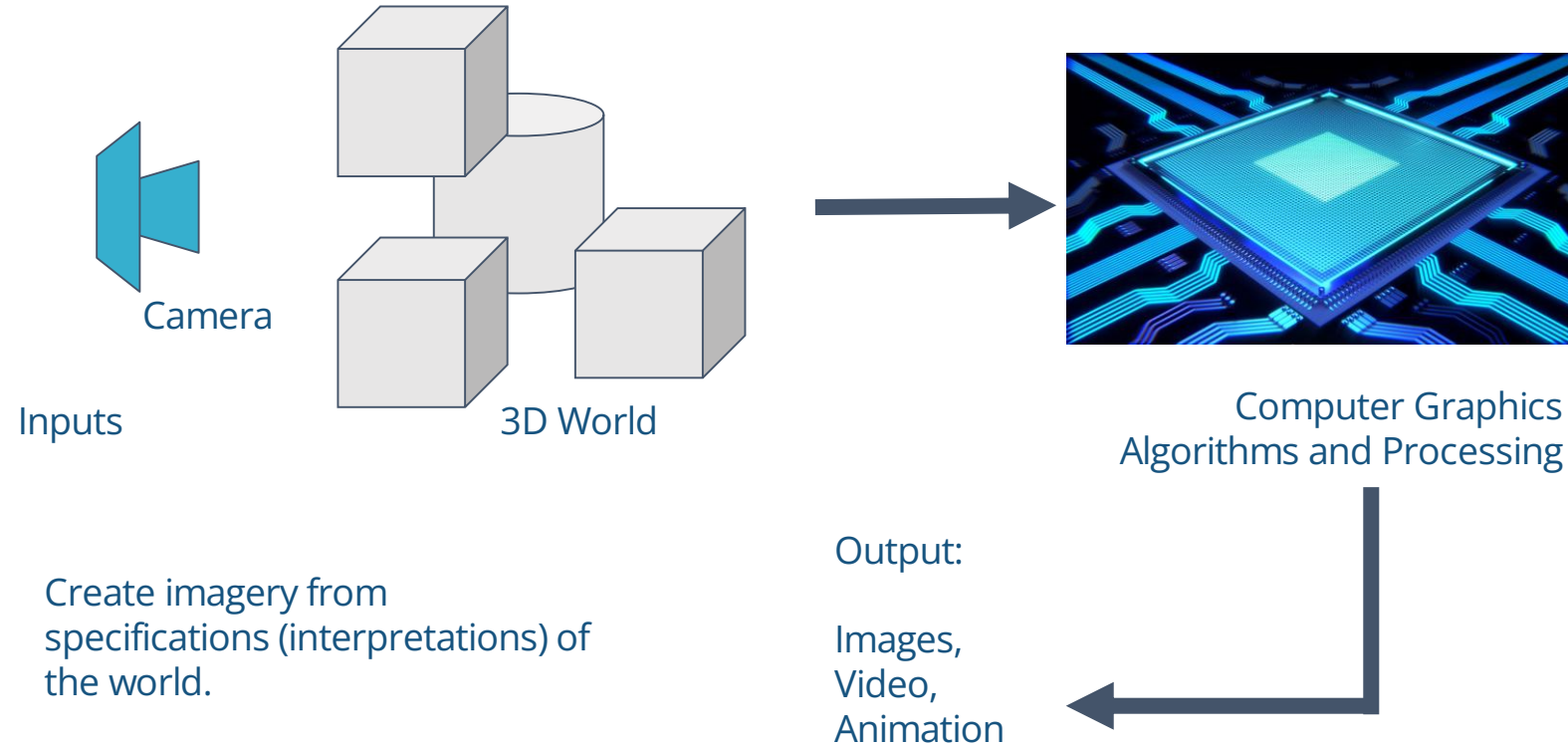
What is Computer Graphics?

- CG is the *art* and *science* of using the computer to make images.
- Study of methods (artistic, mathematical, algorithmic, software) and systems (mechanical, electronic, hardware) to create, control and manipulate pictorial data on the computer.

What is Computer Graphics?

- Forget the definitions!
- Remember **why** you are here.
 - Why are you here - listening to this lecture?

What is Computer Graphics



Impact of Computer Graphics

- Read a bit of history
 - https://en.wikipedia.org/wiki/Computer_graphics
- Among other things, advances in CG
 - Caused computer hardware development to accelerate - think GPUs
 - Made better real-time algorithms possible - networking, files systems, simulations, multi-threading
 - Enabled modern machine learning

Course Content

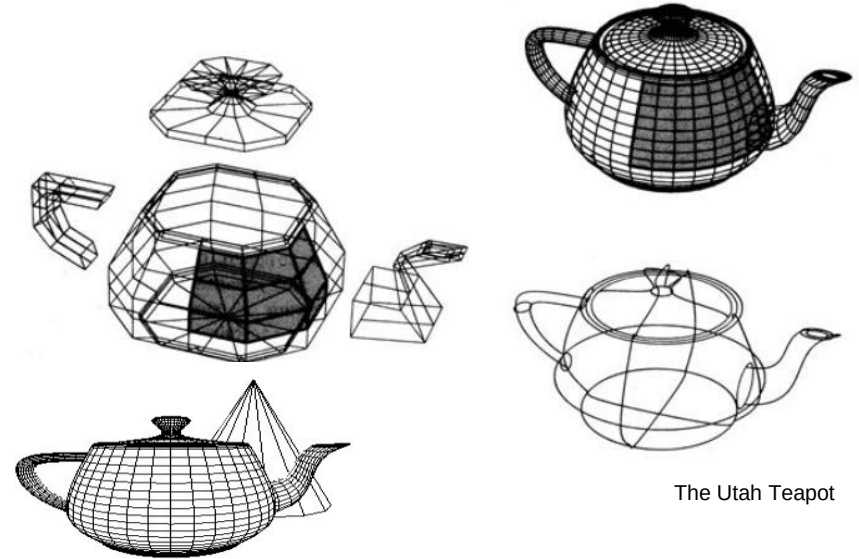
Three parts of Computer Graphics

Three parts of Computer Graphics

- Modelling - Creating the synthetic virtual world
 - Creating and placing objects
 - Creating and placing a camera
- Rendering - Drawing the synthetic virtual world
 - Modelling Illumination Transfer and Shading
 - Visible/Hidden surface determination
 - Online/Real-time and Offline
- Animation - Moving the synthetic virtual world
 - Keyframing and Interpolation

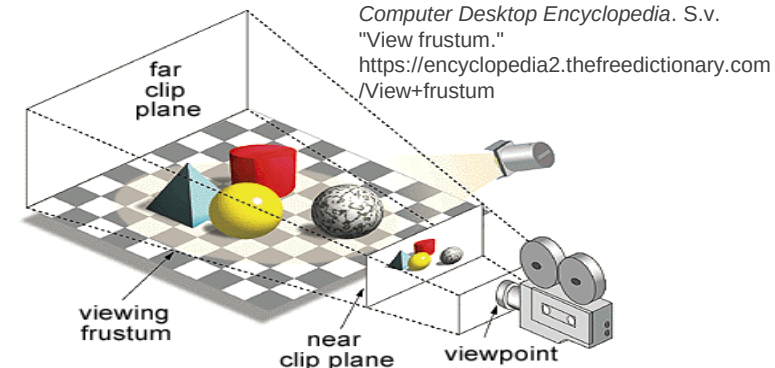
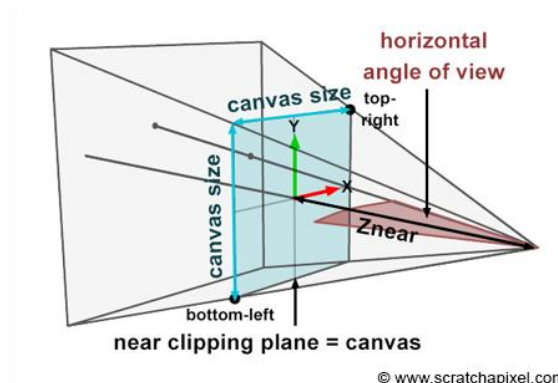
Modelling and Viewing

- Lines, Polygons
- Curves, Surfaces
- Modelling transformations
- Hidden Surfaces
- Viewing transformations



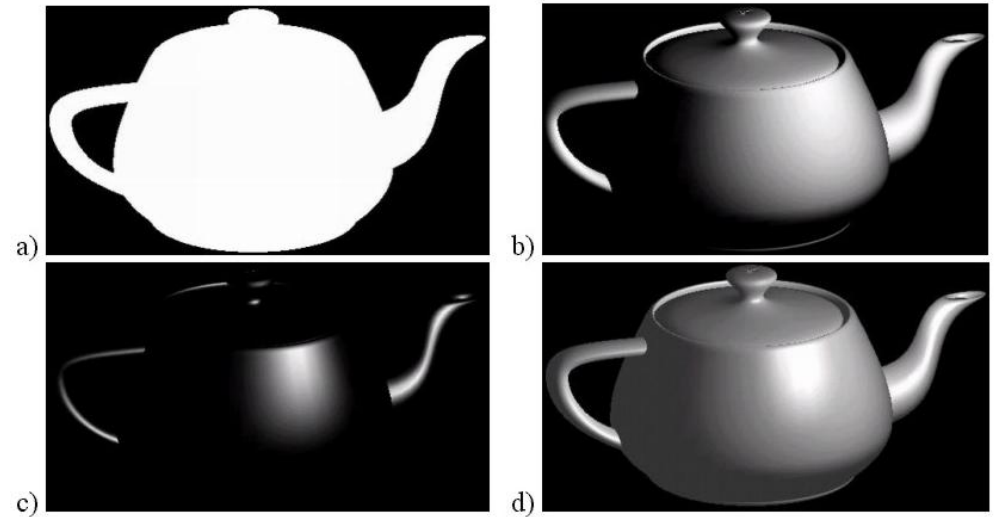
The Utah Teapot

<https://marier.design/work/live/teapot/history>



Rendering

- Shading
- Lighting models
- Texture mapping
- Ray Tracing



<http://www.c-jump.com/bcc/>

Created by Gilles Tran with POV-Ray 3.6 using Radiosity

https://renderman.pixar.com/resources/RenderMan_20/images/rfkTeapotsRIS.png



Animation

- Transformations
- Interpolation
- Keyframing



Luxo Jr., PIXAR, 1986



Neha and Monica, CS675, 2010



Chirag Sethi and Achin Bansal, CS 475, 2009

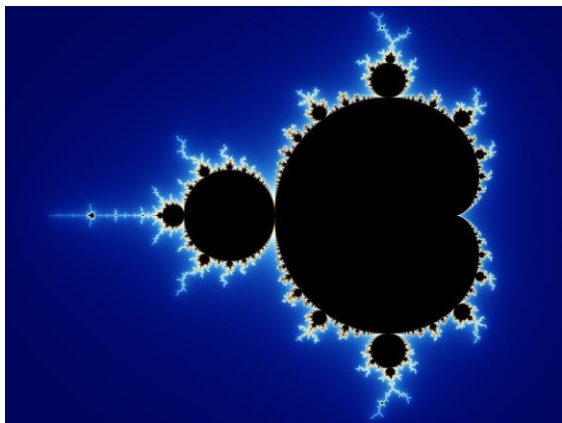


Atul Rokade, CS675, 2012

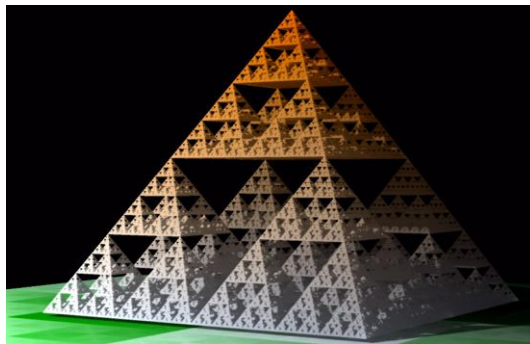


Darshan Kapashi and Shivam Agarwal, CS475, 2011

Drawing in 2D (and 3D)



Created by Wolfgang Beyer with the program Ultra Fractal 3. -
Own work, CC BY-SA 3.0,
<https://commons.wikimedia.org/w/index.php?curid=321973>

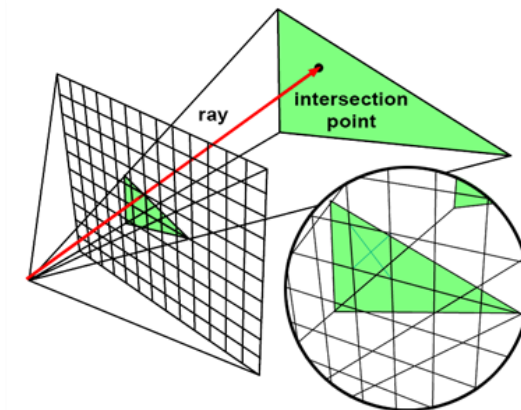


https://en.wikipedia.org/wiki/File:Sierpinski_pyramid.jpg



https://autodesk.blogs.com/between_the_lines/2011/01/the-infamous-utah-teapot-and-its-role-played-in-3d-graphics-history.html

Colouring Pixels - Rasterization

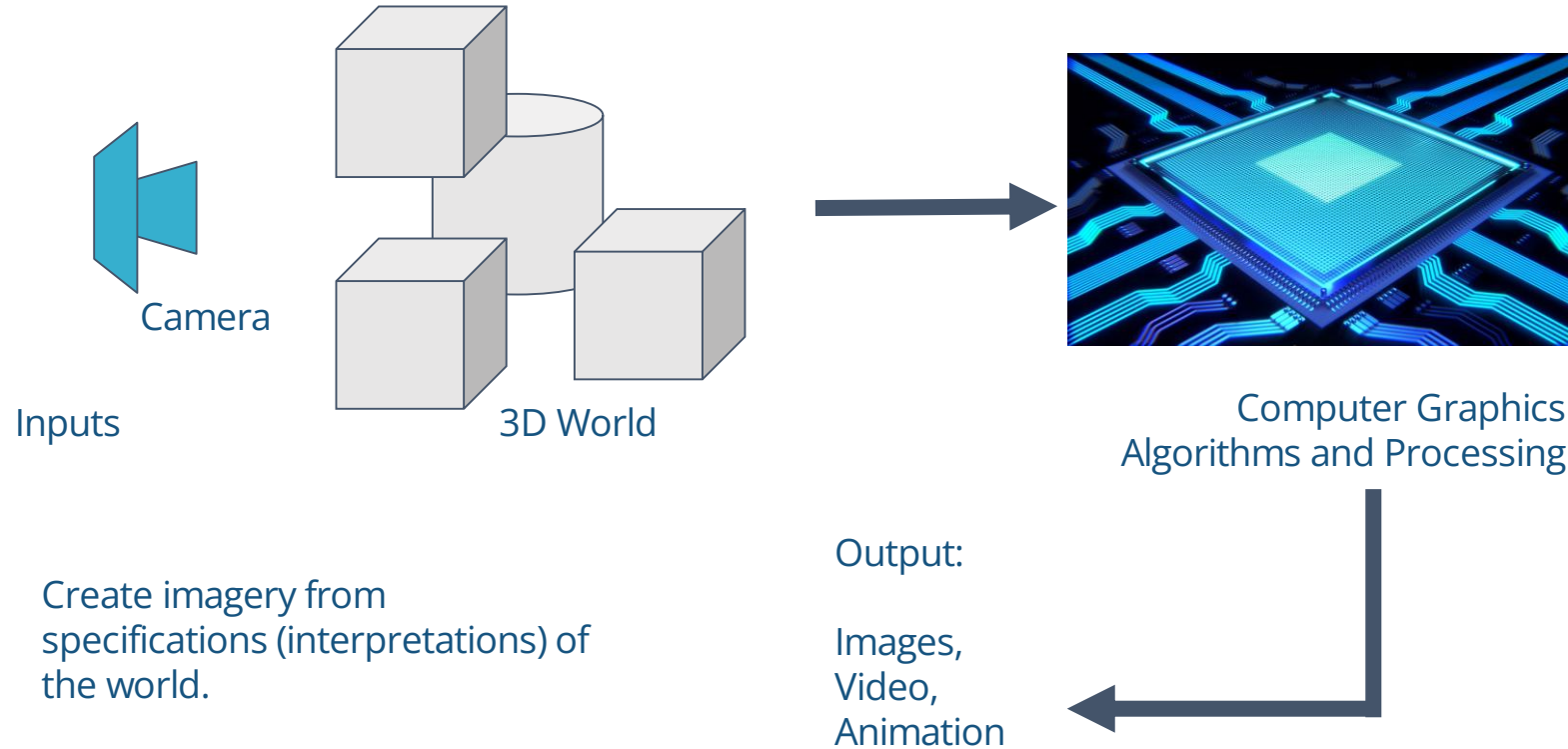


© www.scratchapixel.com

Visual Computing

How does CG relate to other areas in Visual Computing?

Computer Graphics

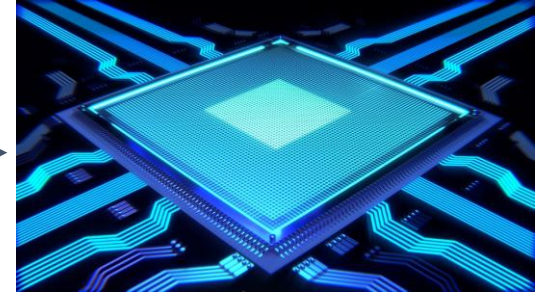


Computer Vision



Input Image or Video

Interpret the world
from Images/Video



Computer Vision Algorithms
and Processing

Output:

Segmentation,
Classification,
Recognition,
Tracking,
Reconstruction



Image Processing

- Processing the images to make them amenable for higher level algorithms
 - Sometimes called low level Vision
 - Blur, Sharpen, Adjust contrast, Equalize..
- These distinctions are fuzzy.

Intersections

Augmented Reality



Pokémon Go, Niantic Inc. 2020

Intersections

VFX



Intersections

Computational Photography



sources/destinations



cloning



seamless cloning

Visual Computing

- Computer Graphics
- Computer Vision
- Image Analysis and Processing
- Computational and Digital Geometry
- Visualization - Scientific, Information
- Virtual, Augmented and Mixed Reality
- Computational Photography

Visual Computing

- Computer Graphics
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-
- Other useful areas: Machine learning, Signal Processing, Perception, Robotics, Games, Imaging, HPC and GPUs, Speech, Cinema, Art ...

Course Logistics

Content sources, Quizzes and Exams, Assignments

Course Details

- Slot 4
 - Tue, Thu : 8:00am – 9:30am
- Classroom – CC 103
- Teaching Assistants
 - Shivang Goswami
 - Kshitij Aphale

Course Details

- Course page

<https://www.cse.iitb.ac.in/~paragc/course/2026-02-cs675/>

- Course material (on Moodle)
 - Videos, Slides, Code Tutorials, Notes and Demos
 - Reading resources, Textbook
- Eligibility and Pre-requisites
 - Data Structures, Linear Algebra

Moodle and Github

- All class material uploaded in IIT Bombay Moodle.
 - Slides, videos
 - References
- All programming assignments on Github Classroom.
 - In teams of two - please find a partner.
 - Collaborative
 - Repository
 - Link to join will be sent with the first assignment.

Other Logistics

- Office hours
 - Talk to me after class
- Online Platform
 - CS675 team on MS Teams
 - Instructions to join will be made available after August 17, 2026.

Other Logistics

- Email Etiquette
 - Email must have a subject line.
 - Email subject must begin with `[CS675]:' followed by the rest of the subject line.
 - Write your emails yourself, instead of copy-pasting from chatgpt.

Assessment - Quizzes and Exams

- We will have at least two graded quizzes in class.
- Aug 26, Oct 14; 5.05pm-6.00pm, Slot XA.
- We will have two exams – mid-semester, end-semester.
- September 12-20, 2026; November 9-21, 2026.
- Exams and quizzes will account for **at least 50%** of the course grade.

Assessment - Programming Assignments

Programming assignments account for **at least 35%** of the course grade.



Assessment - Programming Assignments

- Medium to large programming assignments will be given periodically.
- Submissions will be on Github classroom.
- Links to assignment repositories will be made available for each assignment, on Github classroom.
 - There will be no extensions.

Assessment - Participation

- This is dependent on participation/answering/asking questions in class. Engage with what is happening in the course.
- Just attending the course passively does not guarantee you these marks.
- This will account for **15%** of the course grade.
- **If you have less than 50% attendance, you will not get these marks.** You will get a DX instead.

Plagiarism Policy

Everything that you submit is checked for plagiarism

- Cite sources if you borrow (even if from your classmates/seniors or copy from AI generated code).
- All involved parties get the same penalty.
- At best, you will fail the course.

This will be implemented very strictly.

AI use for programming

For the assignments you submit - I expect you to know what **every single line of code** means

- It is ok to take the help of AI coding agents, but do not let it *replace* your understanding or programming skills.
- Practice writing the code without resorting to AI at the first opportunity.

Attendance Policy

- The DX grade will apply for very low attendance (<50%).
- If you plan to drop *after* the Aug 3 – please tell me.

Unsought Advice

- Attend class – just the slides and book will not be sufficient.
- Do the assignments – plan early, do not ask for extensions.
- If you do not understand something – ASK! Ask early, ask often.
- If what I am teaching seems irrelevant – ask why it is being taught during office hours.
- If you think I am teaching something incorrect – point it out.
 - If you do it often enough – *correctly* – you get a bonus!

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Unsought Advice

- Forget the grade.
- Enjoy the course and have fun learning Computer Graphics.

